

Final Project: Heartbeat Monitor

Section 007

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1 Abstract

Signal processing is key in effective circuit design, and designing a heart beat monitor is an effective way to put that to use. Designing a PPG heart beat monitor starts with an optical sensor. Because of how thick blood is, it can interrupt an infrared wave. This led to the decision to create an infrared emitter and receiver to sense when blood is pumping through the body. This small wave (~ 100 mV), will be amplified through active filters by 100 times, and isolated into one heart beat wave. After that, the signal will go through a hysteresis comparator, where the AC signal will turn into a square wave, or a DC output for the LED. This will light the LED up when the user's heart beats.

2 Introduction

Easily reading a heart rate is important in a plethora of procedures and instances. It can be hard to find a pulse and having the ability to see the heart beating on either an LED or an oscilloscope is important in medicine and health sciences.

One solution to this is creating a PPG to blink an LED with your heart beat. Our design will utilize an infrared emitter and receiver in order to track changes in blood volume through how the infrared light is getting through to the receiver. Each beat will reduce the amount of light getting through, creating an AC output through the phototransistor. This signal can then go through intensive signal analysis, by creating active filters for selecting a range of frequencies and reducing noise. Then we can create an AC to DC converter by utilizing a comparator with hysteresis. This will create a square wave, that will turn on and off our LED, displaying the heartbeat.

3 Equipment

- Analog Discovery 2 (AD2)
- LM324N (op-amp)
- IR204 (infrared emitter)
- PT204-6B (phototransistor)
- LM339 (Comparator)
- LM431 (Adjustable Zener Shunt Diode)
- Assorted Resistors

- Assorted Capacitors
- Assortment of Wires
- NMOS transistor
- LED

4 Procedure

1. Construct each element at a time, testing as each one is built.
2. Construct the optical sensor and measure at v_{PT} . Choose resistor values that make the signal have a v_{p-p} of about 80 mV.
3. Construct an active high and low pass filter that have cutoff frequencies of 40 BPM and 200 BPM respectively [1], in order to only let through the standard heart beat signal. Test each filter extensively and ensure the gain is no more than 1000.
4. Construct a comparator with hysteresis, as seen in the schematic. Adjust resistor values until the threshold is about 1 V.
5. Test this section extensively with different waves.
6. Construct the output with an NMOS transistor as an active load.
7. Construct the voltage divider circuit with an adjustable zener diode. Adjust the resistor value until the output is 2.5 V.
8. Connect each element of the circuit together as seen in the schematic and test extensively.

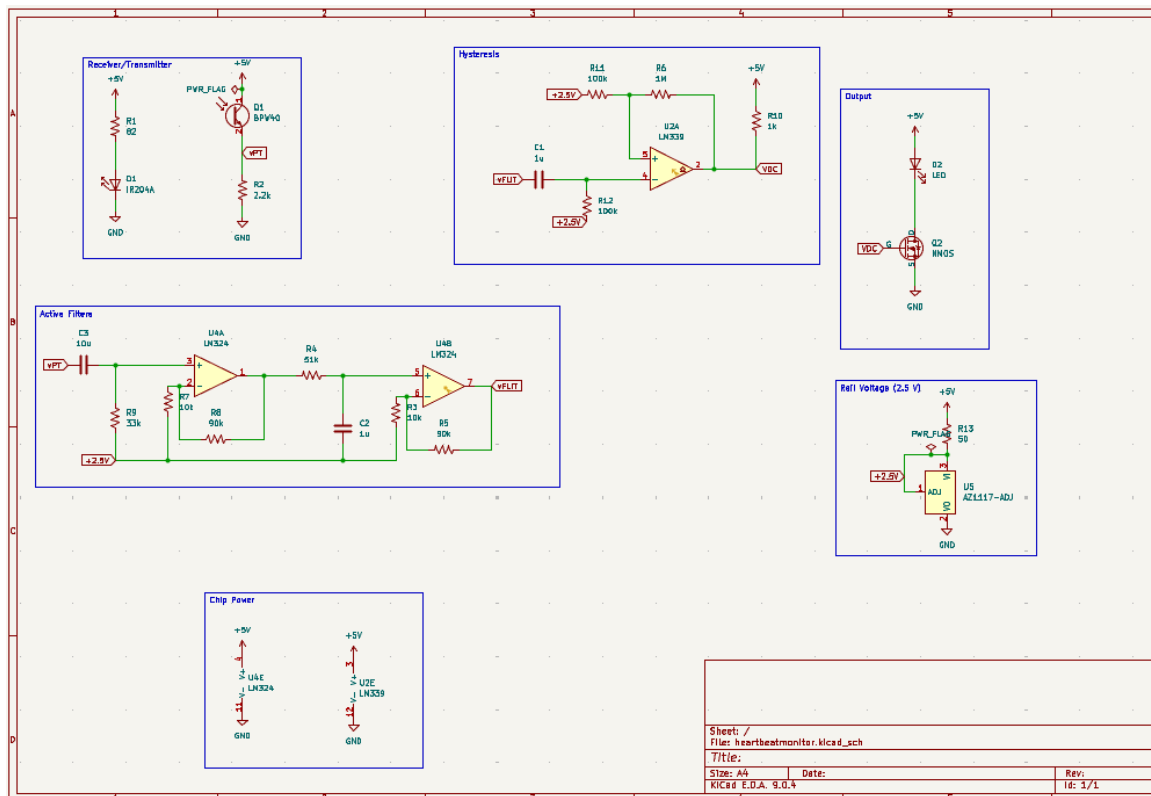


Figure 1: Full Schematic

5 Theory

Optical Sensor

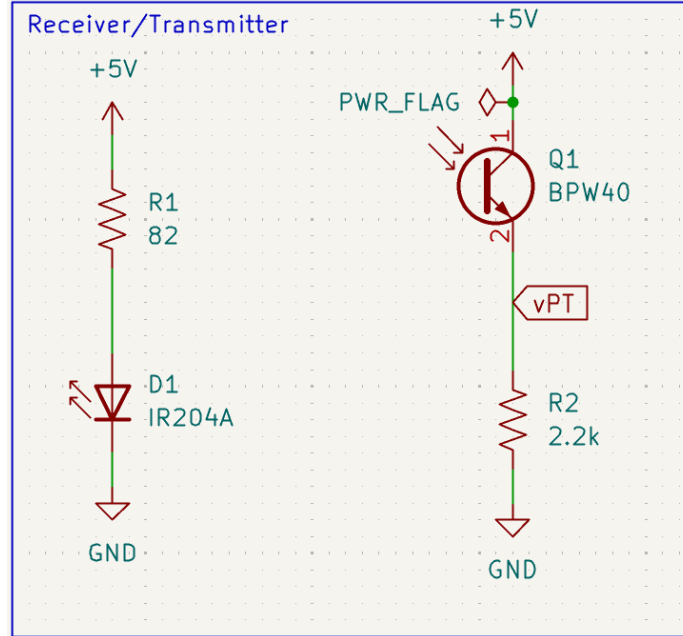


Figure 2: Optical Sensor Schematic

The small changes in emitted infrared light received due to blood flow is causing a small wave output, in v_{PT} . Due to how small the received voltage is, it will require significant amplification. The drain current is calculated as:

$$I_D = \frac{V_{CC} - V_D}{R_D} \quad (1)$$

The PT204-6B operates in the forward region, where current is proportional to the received infrared light intensity. The output voltage is calculated as:

$$v_{PT} = i_{PT} R_E \quad (2)$$

Active Filters

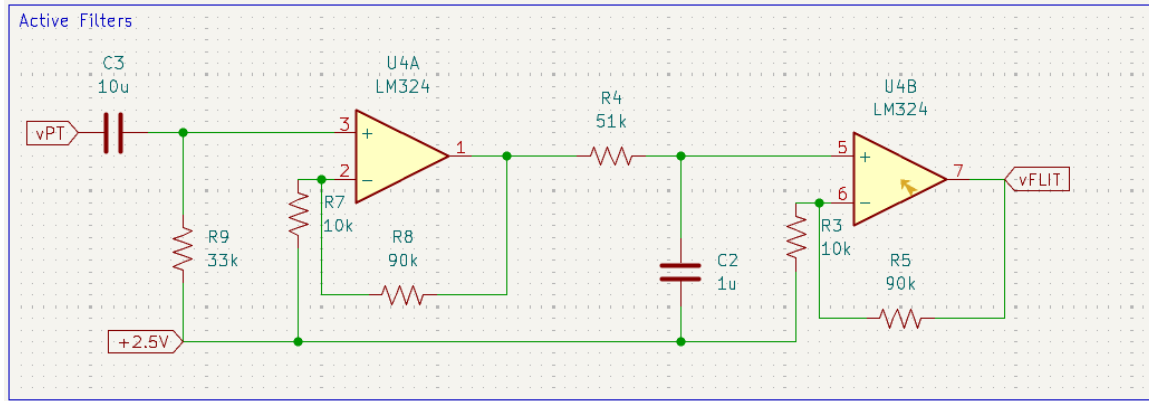


Figure 3: Active Filters Schematic

With the addition of active filters, we can only allow the heartbeat signal through the circuit, while also amplifying it. Since the average heart beat is between 0.67 Hz and 3.33 Hz, this will be our band of frequency we will allow to pass through.

The cutoff frequency of both the high and low pass filter can be calculated using the following equations:

$$\omega_c = \frac{1}{RC} \quad (3)$$

$$f_c = \frac{1}{2\pi RC} \quad (4)$$

The gain of each amplifier was decided to be 10, in order to actually use the very small signal from the heart beat, and so it is not so big, it will cause error when combined.

To calculate resistor values we used the inverting amplifier configuration [2]:

$$A_v = 1 + \frac{R_f}{R_1} \quad (5)$$

Analog to Digital

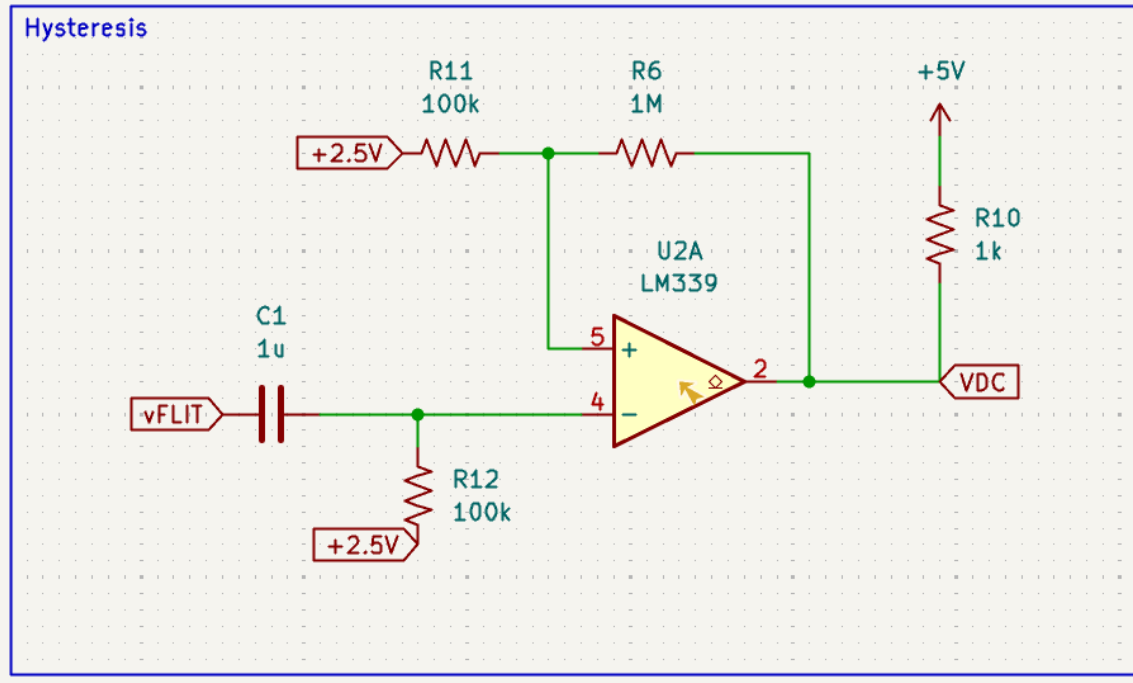


Figure 4: AC to DC Converter Schematic

In order to output a high voltage to power an LED when the sensors detect a heart beat, we needed an element that could output V_{DD} when a signal was detected. This was done with a Schmitt trigger comparator. As this wave is going up, it will hit a threshold voltage that is slightly higher than the mid point, after this, the threshold will reflect across the y-axis, meaning the output will not go back down until the signal is all the way past the negative threshold [3].

To calculate resistor values for the comparator with hysteresis, we used the equation:

$$v_{hyst} = (V_{DD} - V_{SS}) \frac{R_1}{R_1 + R_2} \quad (6)$$

This value must exceed the noise floor of the filtered signal, but remain below the peak-to-peak amplitude of the PPG waveform. To counter this, a high pass filter with a cutoff frequency of 2Hz is placed before the comparator.

$$f_c = \frac{1}{2\pi RC} \quad (7)$$

Reference Change

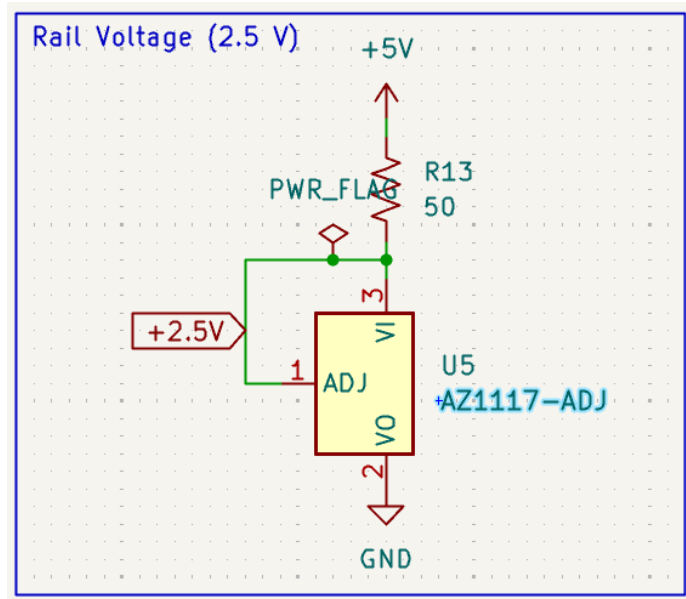


Figure 5: 2.5 Rail Schematic

In order to meet the specifications with only one voltage source, an Adjustable Schunt Zener Diode was placed with a reference voltage of 2.5V. The op-amp and comparator were then grounded so the reference was still halfway between the positive and negative inputs. In order to calculate the resistor value in series with the diode, this equation was used:

$$R_s = \frac{V_{IN} - V_{OUT}}{I_Z - I_{load}} \quad (8)$$

6 Calculations

Optical Sensor

Infrared emitter resistor R_D :

Target $I_D = 46mA$ (< 100 mA max for IR204 [4]), $V_D = 1.2V$, $V_{CC} = 5V$

$$R_D = \frac{V_{CC} - V_D}{I_D} = \frac{5 - 1.2}{0.046} = 82.6 \approx 82\Omega$$

Receiver Resistor R_E :

Phototransistor current $\sim 1 - 3mA$. [5]

$$R_E = \frac{v_{pt}}{i_{pt}} = \frac{1 - 3V}{1 - 3mA} = 1k - 3k\Omega \approx 2.2k\Omega$$

Active Filters

High Pass Filter

Filtering:

Target BPM = 40 BPM $f_c = \frac{BPM}{60} = 0.667Hz$
Target $f_c < 0.667$ Hz. Chosen $R_3 = 33k\Omega$, $C = 10\mu F$.

$$f_c = \frac{1}{2\pi R_3 C} = \frac{1}{2\pi * 33 \times 10^3 * 10 \times 10^{-6}} = 0.482Hz$$

Gain:

Target Gain: 10

$$A_{vh} = 1 + \frac{R_f}{R_1} = 10$$

Chosen $R_f = 90k\Omega$, $R_1 = 1k$.

Low Pass Filter

Filtering:

Target BPM = 200 BPM $f_c = \frac{BPM}{60} = 3.33Hz$
Target $f_c > 3.33$ Hz. Chosen $R_3 = 33k\Omega$, $C = 1\mu F$.

$$f_c = \frac{1}{2\pi R_3 C} = \frac{1}{2\pi * 33 \times 10^3 * 1 \times 10^{-6}} = 4.82Hz$$

Gain:

Target Gain: 10

$$A_{vl} = 1 + \frac{R_f}{R_1} = 10$$

Chosen $R_f = 90k\Omega$, $R_1 = 1k$.

Total Band Pass Gain:

$$A_v = A_{vh} \times A_{vl} = 10 \times 10 = 100$$

Analog to Digital

Comparator with Hysteresis Design:

$v_{hyst} \sim 1V$ for 10 V range and ~ 0.5 for 5 V range.

Chosen $R_1 = 100k\Omega$, $R_2 = 1M\Omega$

$V_{DD} = 5$, $V_{SS} = -5V$ or $0V$

With Two Voltage Sources:

$$v_{hyst} = (5 - (-5)) \times \frac{100k\Omega}{100k\Omega + 1M\Omega} = 10 \times 0.091 = 0.91V$$

With One Voltage Source:

$$v_{hyst} = (5 - 0) \times \frac{100k\Omega}{100k\Omega + 1M\Omega} = 5 \times 0.091 = 0.455V$$

Noise Reducing High Pass Filter Design:

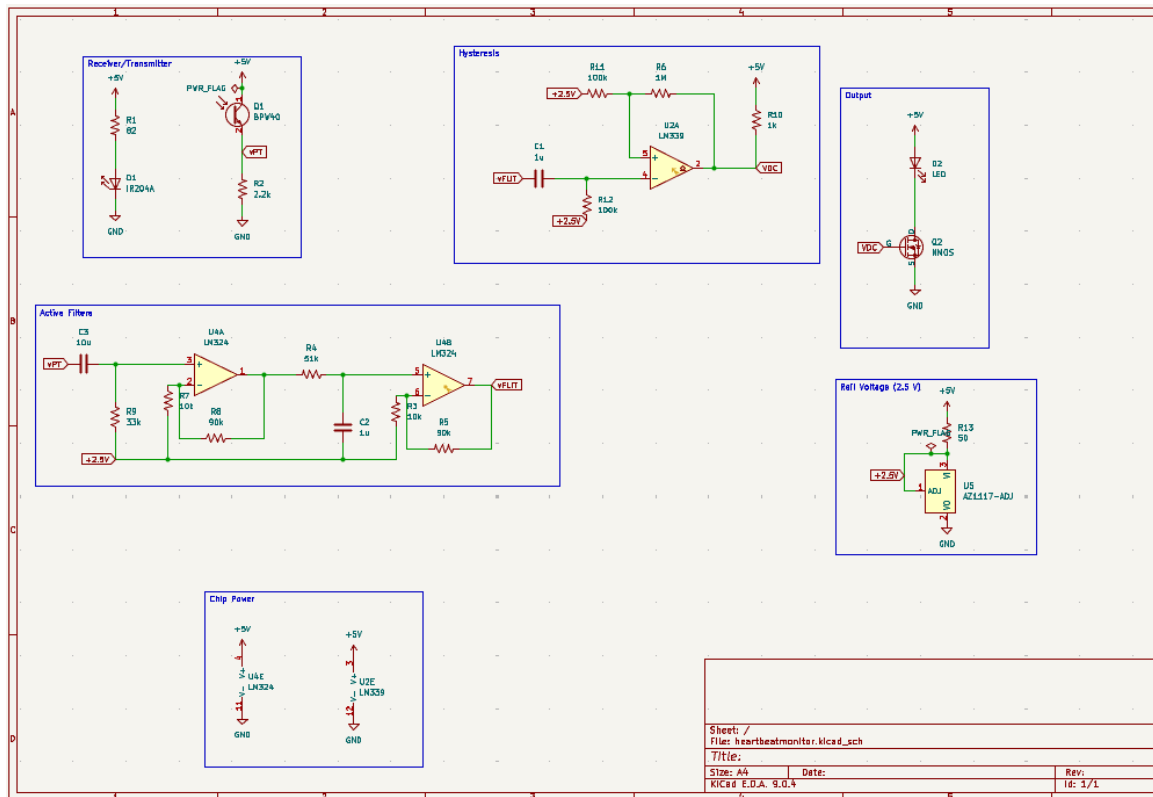
Chosen $R_3 = 100k\Omega$, $C = 10\mu F$

$$f_c = \frac{1}{2\pi R_3 C} = \frac{1}{2\pi * 100 \times 10^3 * 10 \times 10^{-6}} = 0.16Hz \rightarrow (< 2Hz)$$

Reference Change

$$R_S = \frac{V_{IN} - V_{OUT}}{I_Z - I_{load}} = \frac{5V - 0}{0.1A} \approx 50\Omega$$

7 Results



Optical Sensor

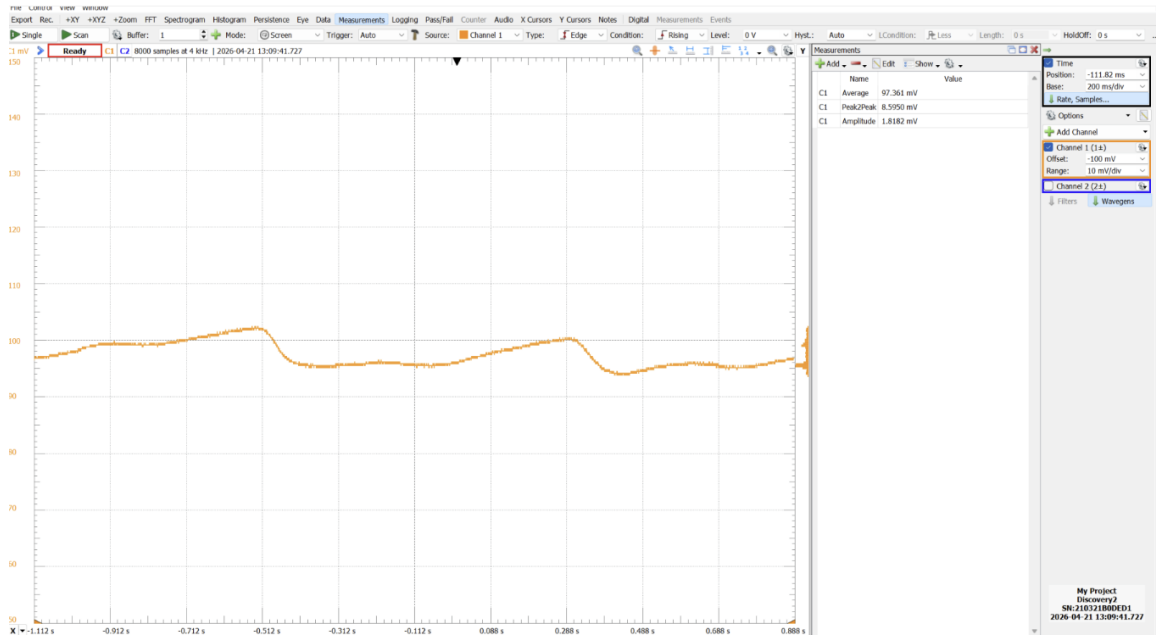
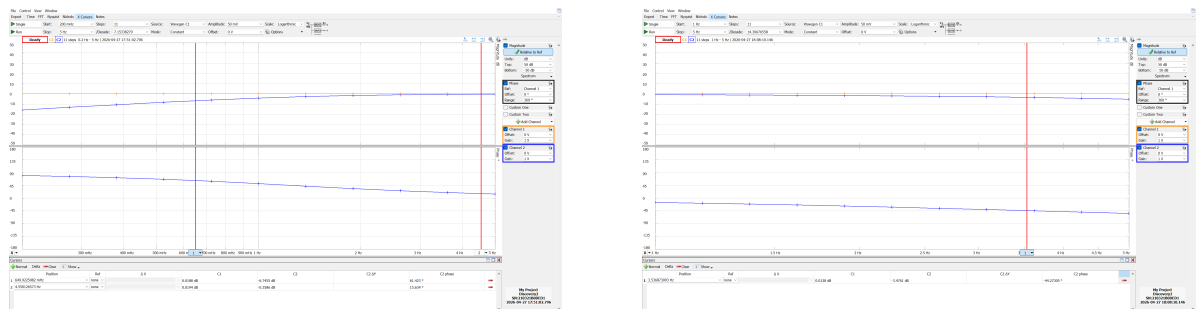


Figure 7: Phototransistor Output (Heartbeat)

The average voltage is about 97mV and the AC ripple amplitude is about 1.8 mV. This is a good value because it will allow us to amplify the signal by 100, as per our op-amp design, and still have a reasonable output voltage. The AC ripple amplitude is about 1.8 mV.

Active Filters



(a) Low Pass Filter

(b) High Pass Filter

Figure 8: High and Low Pass Filter

Parameter	Calculated	Experimental	Percent Error
High Cutoff Frequency (Hz)	0.667	0.649	2.7%
Low Cutoff Frequency (Hz)	3.333	3.537	6.1%

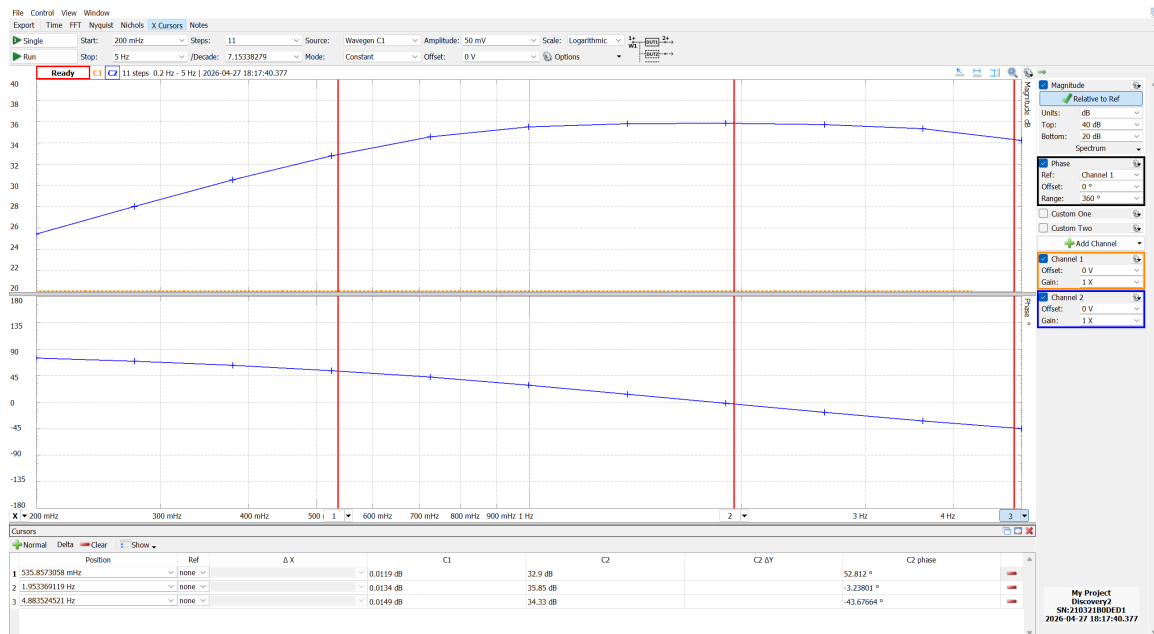


Figure 9: Band Pass Filter

When testing the band pass filter, we noticed a strange pattern. The -3dB points were consistently scaled to make the band pass wider. The new experimental values showed a range of 0.535 Hz to 4.88 Hz. This was very interesting because our high and low pass filters were working as intended. After solving the transfer functions of the filter circuits, we realized that if the gain is being multiplied, then the transfer functions are too. This means that the center frequency amplitude, would realistically be lower than the point where the high and low pass filter cross. This causes the -3dB points to be scaled in either direction, making the filter pass more frequencies than intended. This is adequate for our case because letting in a slightly wider range of frequencies could be helpful in detecting a small heart beat, but it would be accounted for in other scenarios.

Analog to Digital

Hysteresis With Two Voltage Sources:

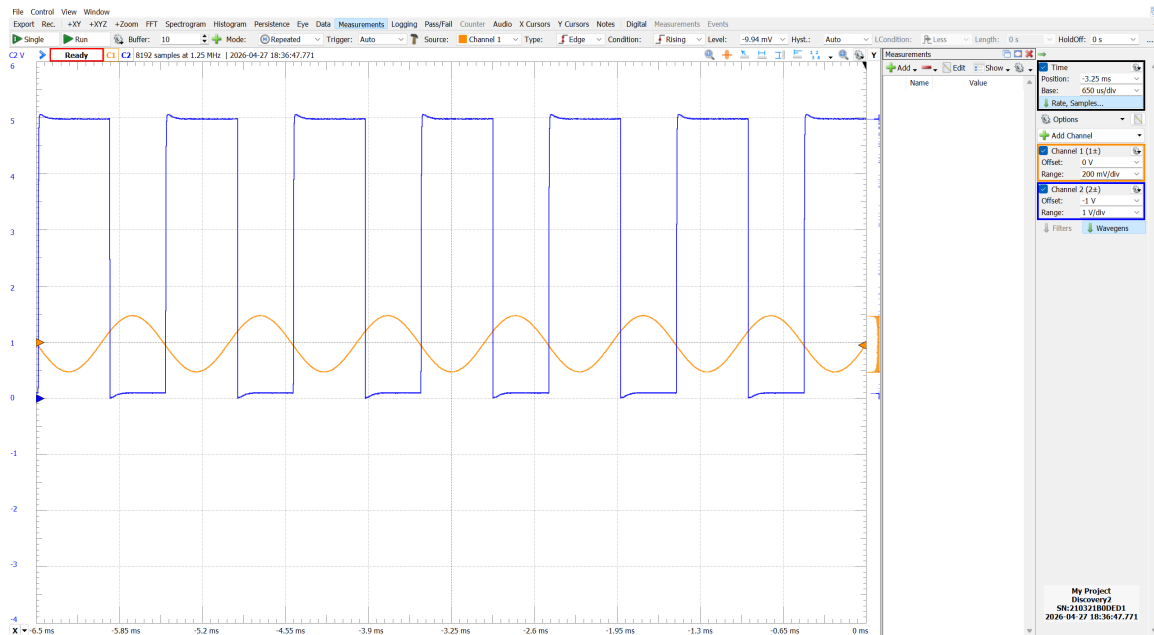
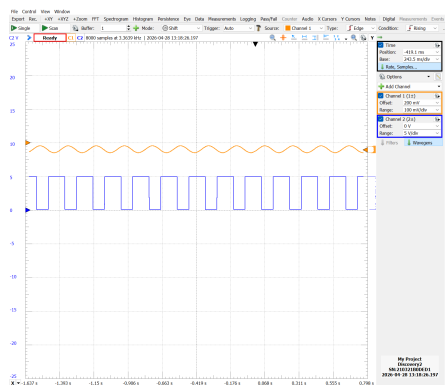


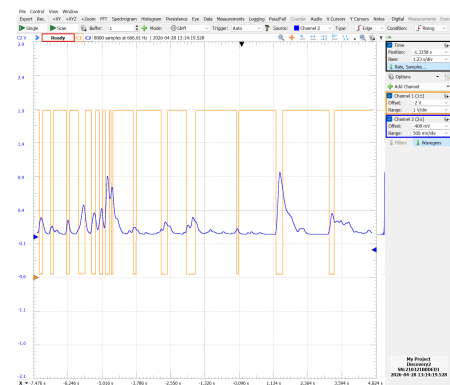
Figure 10: Comparator Output With Sine Wave Input

Calculated Hysteresis Voltage: 0.9 V

Hysteresis With One Voltage Source:



(a) v_{out} and Sine Wave Input



(b) v_{out} and Heart Beat Input

Figure 11: Comparator Output with One 5V Source

Reference Change

Adjustable Zener Diode Output: 2.445 V

$$\text{Percent Error: } \frac{2.5 - 2.445}{2.5} \times 100 = 2.2\%$$

Fully Integrated System:

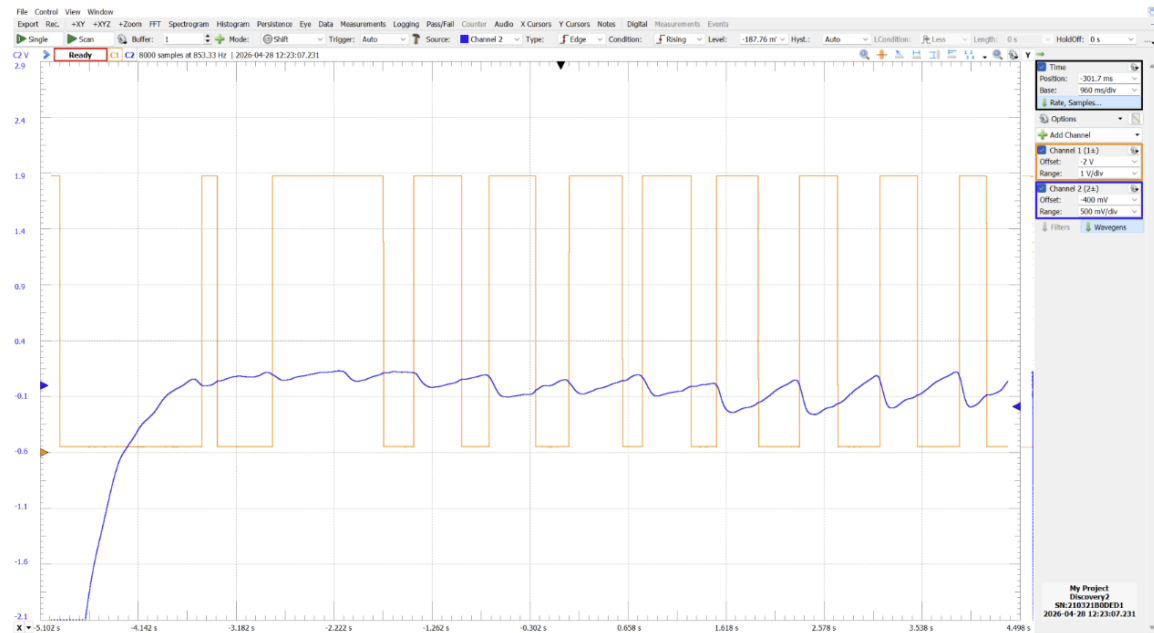
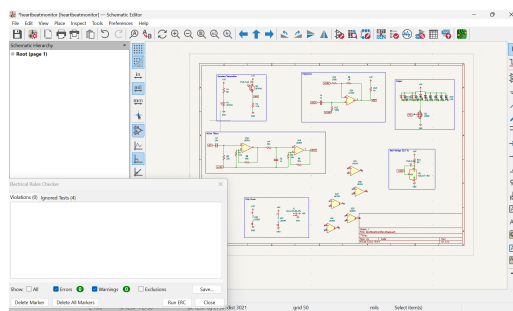


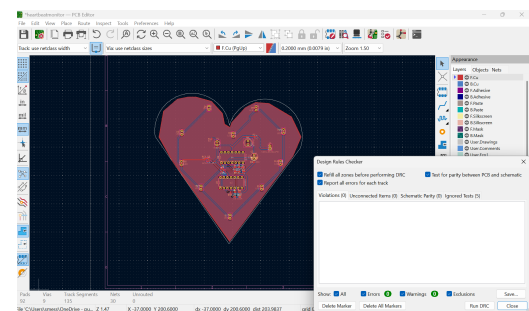
Figure 12: Full Oscilloscope With Heart Beat Input

8 PCB Design:

In order to further reduce noise, I designed a custom PCB for this project.



(a) Electric Rules Checker and Final Schematic



(b) Design Rules Checker and Final PCB Design

Figure 13: Passing Rules Checkers

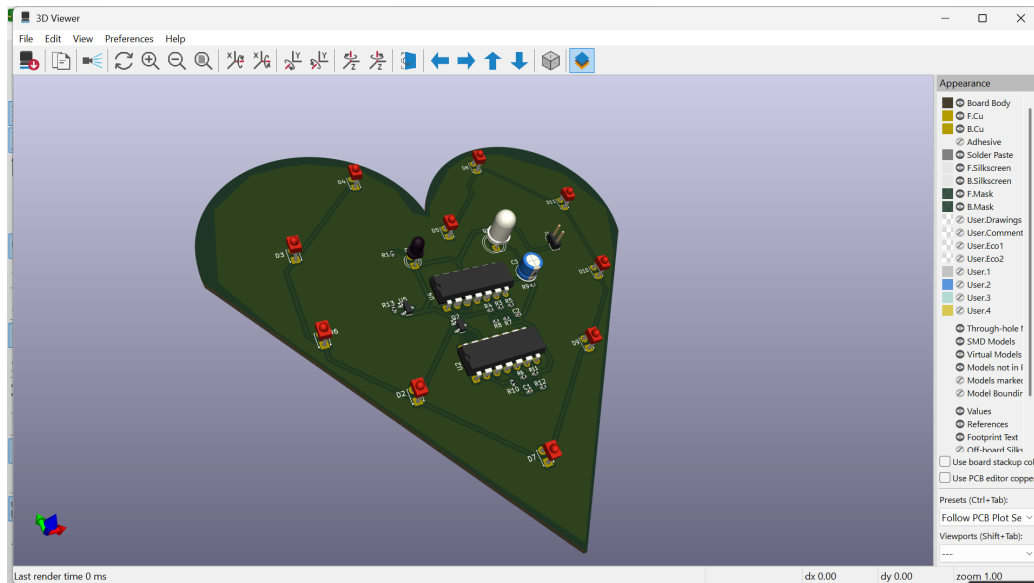


Figure 14: Final 3D Render

9 Conclusion

This sensor successfully detected and displayed the users heart beat. It took the input through the interference of waves through the finger, by detecting the change in blood volume. The signal was then amplified and filtered through active high and low pass filters. Then the signal goes through a comparator with hysteresis, where it turns into a DC signal that can be read by the LED. This system works as expected with minimal error. Because the band pass filter is only there to capture a range of frequencies that could be heartbeats, the band pass was constructed to be slightly wider than the specifications. This was done to ensure all heartbeats will show up at the LED. All specifications provided for the project were met, and any error can be attached to tolerance in resistors, capacitors, wires, and the AD2.

References

- [1] *Purdue University ECE 20008 Lab Manual, Experiment 14: Optical Heart Rate Sensor*, Purdue University, Spring 2022.
- [2] *LMx24-N, LM2902-N Low-Power Quad Op Amps*, Texas Instruments, Jan 2015, available: <http://www.ti.com/lit/ds/symlink/lm324-n.pdf>.
- [3] *LM339, LM239, LM139, LM2901 Quad Differential Comparators*, Texas Instruments, Nov 2018, available: <http://www.ti.com/lit/ds/symlink/lm2901.pdf>.
- [4] *3mm Infrared LED IR204-A*, Everlight, Nov 2016, available: <http://www.everlight.com/file/ProductFile/IR204-A.pdf>.
- [5] *3mm Phototransistor T-1 PT204-6B*, Everlight, Jun 2013, available: <http://www.everlight.com/file/ProductFile/PT204-6B.pdf>.